Gender Classification Model Card

Hackathon for Gender Equity 2.0

1. Model Overview

This model performs binary gender classification from facial images.

Output labels:

- **0 → Male**
- **1 → Female**

The model is implemented in PyTorch and optimized for CPU-only inference, ensuring compatibility with the competition's standardized offline evaluation environment.

2. Dataset(s) Used

The model was trained using a combination of:

- **UTKFace Dataset**
- **FairFace Dataset (Training Split Only)**

Total training samples after cleaning and preprocessing:

- ~42,967 training images
- ~10,741 validation images

The datasets were cleaned to remove:

- Corrupted images
- Extremely small resolution images
- Non-facial images

An 80/20 train-validation split was used.

3. Model Architecture

- Backbone: **ResNet18**
- Pretrained on: **ImageNet**
- Modified final fully connected layer:
    - Input features → 2 output classes

Total Parameters: ~11 million
Trainable Parameters (after fine-tuning): ~11 million

The model was fine-tuned for binary classification.

4. Training Strategy

Training was conducted in two stages:

Stage 1:

- Frozen backbone
- Trained final classification layer
- Learning rate: 1e-4

Stage 2 (Fine-Tuning):

- Entire model unfrozen
- Learning rate: 1e-5

Data augmentation included:

- Random horizontal flip
- Random rotation
- Resize to 224×224

Loss Function:

- CrossEntropyLoss with class weighting to address minor class imbalance

Optimizer:

- Adam

Best Validation Accuracy Achieved:
~89.49%

5. Inference Design

The model:

- Runs on CPU only
- Uses fixed image resizing (224×224)
- Produces softmax probabilities

- Returns predicted label and confidence score (0–1)

No internet access or external services are required.

Inference time per image on CPU is efficient and suitable for real-time use.

6. Performance

Validation Accuracy: ~89.5%
The model demonstrates stable generalization on unseen validation data.

The model is lightweight and optimized for balance between:

- Accuracy
- Speed
- Model size
- CPU compatibility

7. Ethical and Bias Considerations

- The model was trained on publicly available datasets containing demographic diversity.
- However, bias may still exist due to dataset distribution limitations.
- Gender classification based solely on facial appearance does not account for gender identity.
- The model should be used for academic or benchmarking purposes only, not for sensitive decision-making applications.

8. Limitations

- Performance may reduce on:
    - Low-resolution images
    - Extreme pose angles
    - Occluded faces
    - Images with heavy lighting variation
- Binary classification does not represent non-binary gender identities.

Future improvements may include:

- Improved robustness augmentation

- Larger balanced dataset
- More lightweight architecture for smaller model size